PES UNIVERSITY

Department of Computer Science & Engineering

Object Oriented Analysis and Design

UE23CS352B

Assignment 2 Submission

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Section	G
Department	CSE
Campus	RR

Object Oriented Analysis and Design (UE23CS352B) - Lab 2

Case Study on Class Modeling

Case Study: Smart Water Distribution Management System

Classes:

1. User
    a. Attributes:
        i. userID : String
        ii. name : String
        iii. age : Integer
        iv. email : String
        v. contact no : Integer
        vi. passwd : String
        vii. role : UserRole
    b. Methods:
        i. + authenticate(password): Boolean
        ii. + updateProfile(details): void
        iii. + logout(): void
2. City officials
    a. Attributes:
        i. designation : String
    b. Methods:
        i. + defineTariff() : void
        ii. + updateTariff() : void
        iii. + createPolicy() : void
        iv. + monitorPerformance() : void
3. Consumer
    a. Attributes:
        i. consumerID: String
    b. Methods:
        i. + viewConsumption() : List
        ii. + viewHist() : List
        iii. + makePayment(payment : Payment) : Boolean
4. Maintenance Staff
    a. Attributes:
        i. staffID: String
        ii. staffType: String
    b. Methods:
        i. + viewTasks() : List
        ii. + updateTasks() : void
5. Distribution Zone
    a. Attributes:
        i. zoneID: String
        ii. zoneName(): String
    b. Methods:
        i. + listMeters() : List
        ii. + listIssues() : List

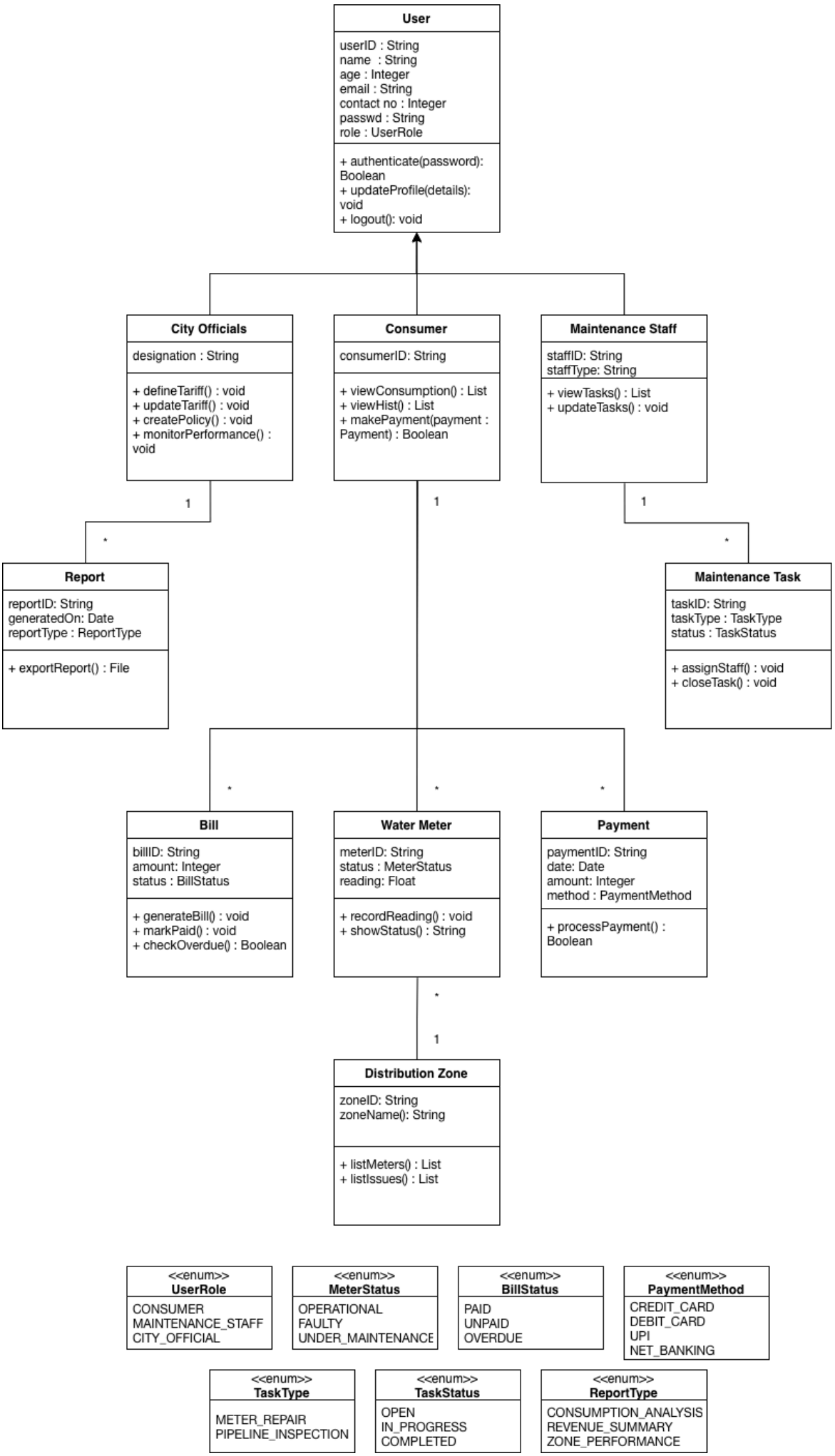
6. Water Meter
    a. Attributes:
        i. meterID: String
        ii. status : MeterStatus
        iii. reading: Float
    b. Methods:
        i. + recordReading() : void
        ii. + showStatus() : String
7. Bill
    a. Attributes:
        i. billID: String
        ii. amount: Integer
        iii. status : BillStatus
    b. Methods:
        i. + generateBill() : void
        ii. + markPaid() : void
        iii. + checkOverdue() : Boolean
8. Payment
    a. Attributes:
        i. paymentID: String
        ii. date: Date
        iii. amount: Integer
        iv. method : PaymentMethod
    b. Methods:
        i. + processPayment() : Boolean
9. Maintenance Task
    a. Attributes:
        i. taskID: String
        ii. taskType : TaskType
        iii. status : TaskStatus
    b. Methods:
        i. + assignStaff() : void
        ii. + closeTask() : void
10. Report
    a. Attributes:
        i. reportID: String
        ii. generatedOn: Date
        iii. reportType : ReportType
    b. Methods:
        i. + exportReport() : File

Enums:

11. UserRole
    a. CONSUMER
    b. MAINTENANCE_STAFF

    c. CITY_OFFICIAL
12. MeterStatus
    a. OPERATIONAL
    b. FAULTY
    c. UNDER_MAINTENANCE
13. BillStatus
    a. PAID
    b. UNPAID
    c. OVERDUE
14. PaymentMethod
    a. CREDIT_CARD
    b. DEBIT_CARD
    c. UPI
    d. NET_BANKING
15. TaskType
    a. METER_REPAIR
    b. PIPELINE_INSPECTION
16. TaskStatus
    a. OPEN
    b. IN_PROGRESS
    c. COMPLETED
17. ReportType
    a. CONSUMPTION_ANALYSIS
    b. REVENUE_SUMMARY
    c. ZONE_PERFORMANCE

Use Case Diagram:


User
userID : String
name : String
age : Integer
email : String
contact no : Integer
passwd : String
role : UserRole
+ authenticate(password): Boolean
+ updateProfile(details): void
+ logout(): void
City Officials
designation : String
+ defineTariff() : void
+ updateTariff() : void
+ createPolicy() : void
+ monitorPerformance() : void
Consumer
consumerID: String
+ viewConsumption() : List
+ viewHist() : List
+ makePayment(payment : Payment) : Boolean
Maintenance Staff
staffID: String
staffType: String
+ viewTasks() : List
+ updateTasks() : void
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*
1
*
1
*
Report
reportID: String
generatedOn: Date
reportType : ReportType
+ exportReport() : File
Maintenance Task
taskID: String
taskType : TaskType
status : TaskStatus
+ assignStaff() : void
+ closeTask() : void
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Bill
billID: String
amount: Integer
status : BillStatus
+ generateBill() : void
+ markPaid() : void
+ checkOverdue() : Boolean
Water Meter
meterID: String
status : MeterStatus
reading: Float
+ recordReading() : void
+ showStatus() : String
Payment
paymentID: String
date: Date
amount: Integer
method : PaymentMethod
+ processPayment() : Boolean
*
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Distribution Zone
zoneID: String
zoneName(): String
+ listMeters() : List
+ listIssues() : List
<<enum>>
UserRole
CONSUMER
MAINTENANCE_STAFF
CITY_OFFICIAL
<<enum>>
MeterStatus
OPERATIONAL
FAULTY
UNDER_MAINTENANCE
<<enum>>
BillStatus
PAID
UNPAID
OVERDUE
<<enum>>
PaymentMethod
CREDIT_CARD
DEBIT_CARD
UPI
NET_BANKING
<<enum>>
TaskType
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